

Kognitionspsychologie: Session 1

Introduction

Rui Mata, HS 2026

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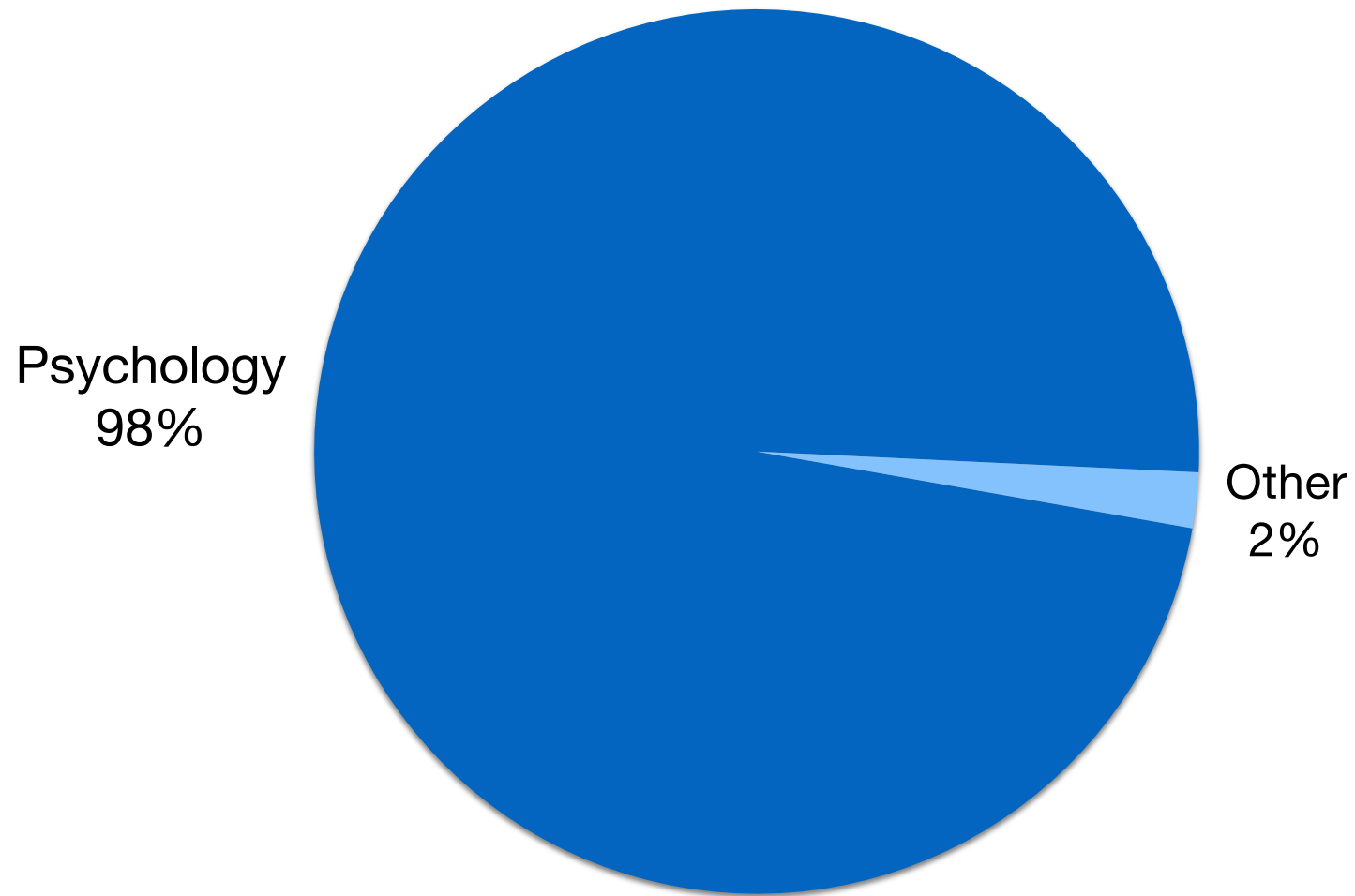
Welcome!



Prof. Dr. Rui Mata

Center for Cognitive and
Decision Sciences (CDS)

You



Learning Objectives for the Semester

You will...

- learn about central theories and models in key areas of psychology
- become familiarized with common methods used in psychology
- learn about examples of applications of psychology to real-world contexts
- reflect about the need for pluralistic explanations in psychology



Pluralistic Explanations

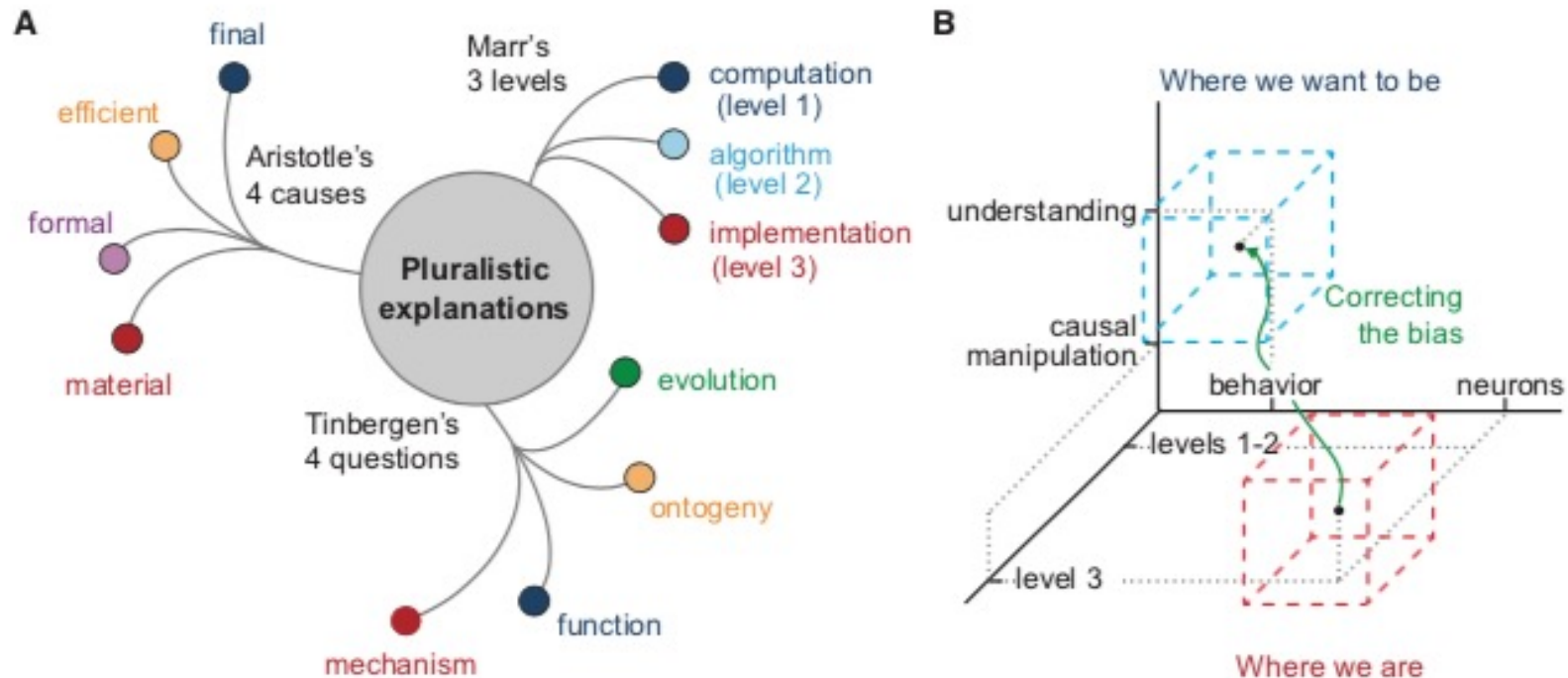


Figure 4. The Future History of Pluralistic Explanation

(A) That understanding of a phenomenon is multidimensional has long been appreciated. Aristotle posited four kinds of explanation: to explain “why” something changes, a polyhedric notion of causality is necessary; one that includes not only the material cause (what it is made out of), but also the other three “whys”: formal (what it is to be), efficient (what produces it), and final (what it is for). Tinbergen also devised four questions about behavior: to go beyond its proximate causation (mechanism) to also considering its evolution, development, and real-world function. Marr’s three levels are also shown.

(B) Three-dimensional space with axes of understanding-manipulation, behavior-neurons, and Marr’s levels. The red box is where we are and the blue is where we should be.

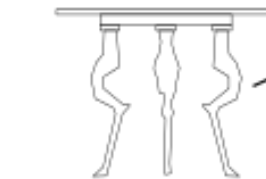
Krakauer, J. W., Ghazanfar, A. A., Gomez-Marín, A., MacIver, M. A., & Poeppel, D. (2017). Neuroscience needs behavior: Correcting a reductionist bias. *Neuron*, 93(3), 480–490. <http://doi.org/10.1016/j.neuron.2016.12.041>

Aristotle's four causes

Material Cause:
Wood



Final Cause:
Dining



Formal Cause:
Design



Efficient Cause:
Carpentry

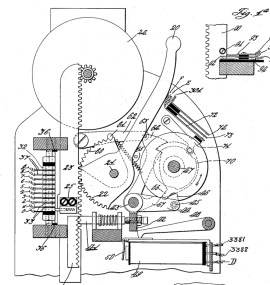
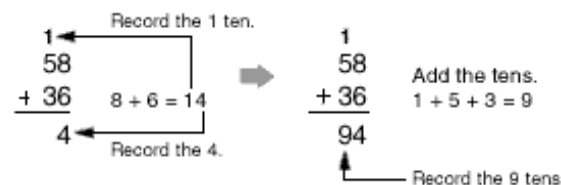
Marr's levels

- **Computational level:** What is the goal of a given process/computation?
- **Algorithmic level:** How can a goal be achieved using a particular set of inputs/outputs, which algorithm describes the required transformations?
- **Implementational level:** How is an algorithm physically implemented (e.g., neural activity)?



Addition

$$S_T = Z_1 + \dots + Z_N$$



David Marr (1945-1980)

Studied mathematics and physiology but later worked as a professor of Psychology at MIT. He integrated results from psychology, artificial intelligence, and neurophysiology to produce a new model of vision. He is particularly famous for proposing a three level view of how to understand information processing systems (see left).

Marr, D. C., & Poggio, T. (1977). From understanding computation to understanding neural circuitry. *Neurosciences Research Program Bulletin*, 15(3), 470–488.

Tinbergen's four questions

Tinbergen argued that there are complementary categories of explanations, involving different kinds and objects of explanation.

FOUR AREAS OF BIOLOGY: FOUR QUESTIONS		Two objects of explanation	
		<u>Developmental/historical</u> A sequence that results in the trait	<u>Single form</u> The trait at one slice in time
Two kinds of explanations	<u>Proximate</u> Explains how organisms work by describing their mechanisms and their ontogeny	<u>Ontogeny</u> Q: How does the trait develop in individuals? A: Description of the trait's forms at sequential life stages, and the mechanisms that control development.	<u>Mechanism</u> Q: What is the structure of the trait; how does it work? A: Description of the trait's anatomy, physiology, regulation, and how the trait works to accomplish a function.
	<u>Evolutionary</u> Explains how a species came to its current form by describing a sequence of forms, and how they were influenced by selection and other evolutionary factors.	<u>Phylogeny</u> Q: What is the phylogenetic history of the trait? A: Description of the history of the trait as reconstructed from its phenotype and genotype precursors	<u>Adaptive significance</u> Q: How have variations in the trait interacted with environments to influence fitness in ways that help to explain the trait's form? A: Description of how variations in the trait have influenced fitness



Nikolaas Tinbergen (1907-1988)

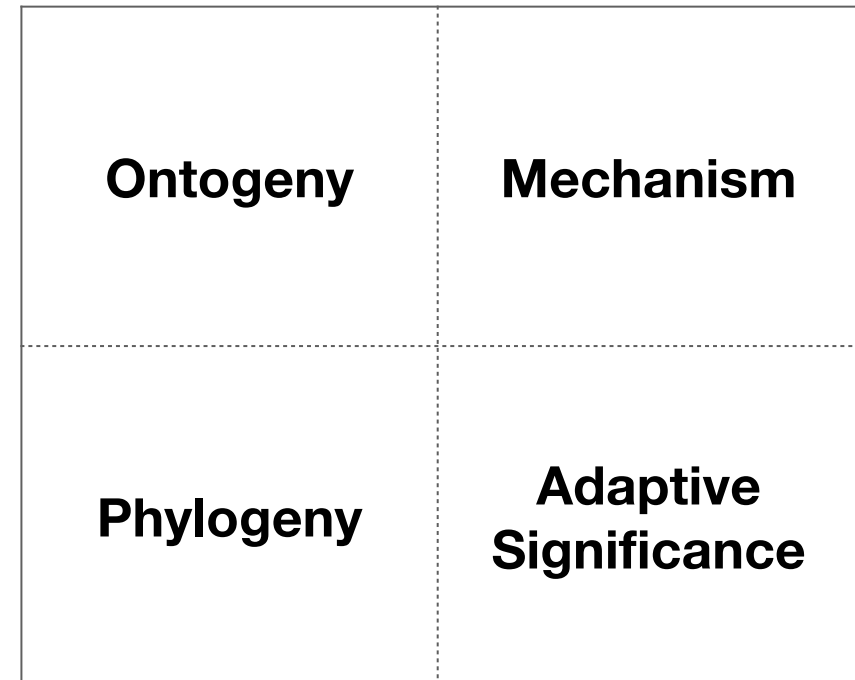
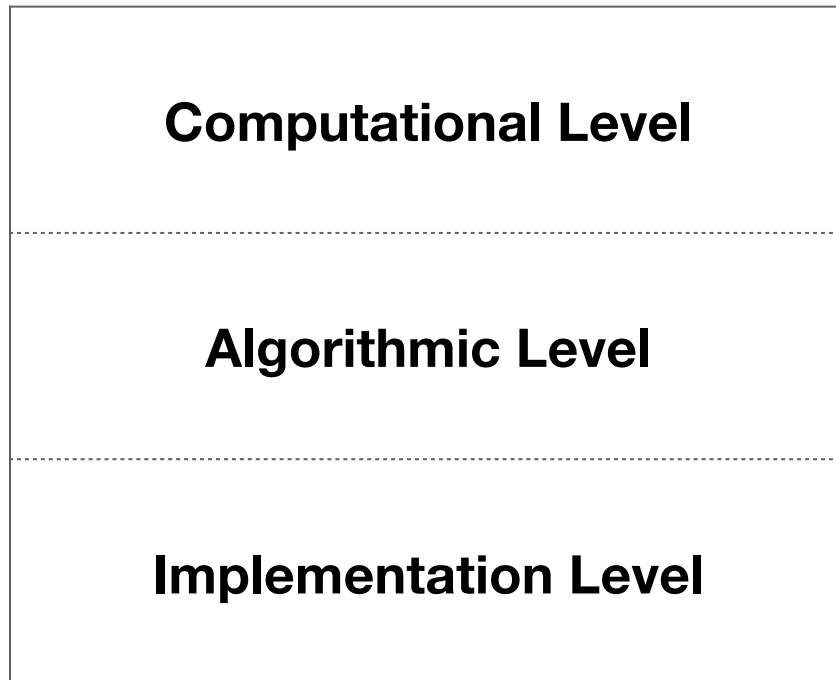
Ethologist, received the Nobel Prize for Medicine in 1973 for “discoveries concerning organization and elicitation of individual and social behaviour patterns”. Tinbergen had a large impact on the field of ethology (i.e., the science of animal behavior) with his book *The Study of Instinct* (1951), in which he proposed to investigate innate behaviour that is not acquired or changed by learning.

Nesse, R. (2013), Tinbergen's four questions, organized: A response to Bateson and Laland. *Trends in Ecology and Evolution*, 28(12), 681-682.

Tinbergen, N. (1963) On aims and methods of ethology. *Z. Tierpsychol*, 20, 410–433.

https://en.wikipedia.org/wiki/Tinbergen%27s_four_questions

Marr's Levels vs. Tinbergen's Questions



Summary

- **Pluralistic explanations:** From Aristotle to Marr and Tinbergen, a recurring idea is that natural phenomena can require several complementary forms of explanation. In psychology, we often want to explain several aspects of a system, such as **what a system does, how it does it, how it develops, how it evolved, and what function it serves.**
- **Marr** organized explanation into **computational, algorithmic, and implementation levels.**
- **Tinbergen** distinguished **mechanism, ontogeny, phylogeny, and adaptive significance.**
- The different perspectives are related but not fully interchangeable, and understanding often comes from combining them rather than reducing one to another.



WHO'S GOT THE ANSWER?

Two theories agree that vision must identify objects despite changes in size and viewpoint. One proposes successive extraction of features; the other assumes matching input directly to stored templates. At which level do the theories mainly disagree? [Select all the correct answers]

☐ A: Computational

☐ B: Algorithmic

☐ C: Implementation

☐ D: None



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Suppose a computer model and a human perform the same task using the same sequence of information-processing steps, but one uses silicon circuits and the other neurons. According to Marr, what differs?
[Select all the correct answers]

• A: Goal

• B: Feelings

• C: Implementation

• D: Computation



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A researcher asks why a memory system should retain some information and discard other when storage is limited. Which answer is closest to Marr's computational level?
[Select all the correct answers]

• A: It prioritizes information useful for future behavior

• B: Items compete during retrieval

• C: Retrieval uses contextual cues

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Four explanations are proposed for seasonal bird migration.
Which two are “proximate” explanations in Tinbergen’s
framework? [Select all the correct answers]

• A: Hormonal changes
triggered by day length

• B: Migration increases winter
survival

• C: Migration evolved from
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• D: Navigation improves with
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A researcher argues that infant crying persists because it elicits caregiving and thereby increased offspring survival. What kind of explanation is being proposed? [Select all the correct answers]

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• C: Phylogeny

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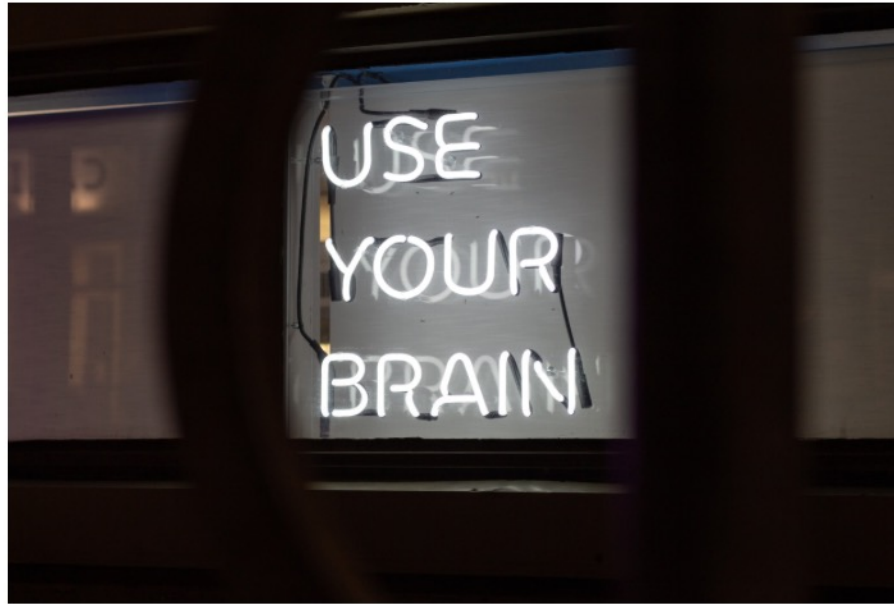
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Course website



Thinking is simply using one's brain! Photo by [Jesse Martini](#) on [Unsplash](#)

Session information

Sessions take place Tuesdays, 10.15–11.45, Pharmazentrum, Hörsaal 1. Slides will be made available shortly before each session.

#	Date	Topic	Slides
1	22.09.2026	Session 1: Introduction	pdf
2	06.10.2026	Session 2: What is intelligence?	pdf
3	13.10.2026	Session 3: Perception	pdf
4	20.10.2026	Session 4: Spatial cognition	pdf
5	03.11.2026	Session 5: Numerical cognition	pdf
6	17.11.2026	Session 6: Language	pdf
7	24.11.2026	Session 7: Memory	pdf
8	01.02.2026	Session 8: Consciousness	pdf
9	08.12.2026	Session 9: Applications	pdf
10	15.12.2026	Session 10: Wrap-up and Q&A	pdf

https://matarui.github.io/KOGPSY_HS26/

WHAT IS INTELLIGENCE?

**And how would one study it based on
Tinbergen's 4 questions?**

